Effects of Spaced Repetition on Long-Term Retention in Undergraduate Biology Courses

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Abstract

This study examines whether spaced repetition scheduling improves long-term retention of course material among first-year undergraduate biology students. Using a randomized controlled design with 214 participants across two semesters, we compared a spaced-repetition study group against a massed-practice control group. Retention was measured through unannounced assessments at one, four, and twelve weeks after instruction.

1. Introduction

Cognitive psychology has long shown that distributing study sessions over time produces stronger memory than concentrating them in a single session, a phenomenon known as the spacing effect. Despite this evidence, most university students rely on cramming before exams, which tends to produce short-lived knowledge that fades within weeks.

Introductory biology is a useful test case because it demands heavy memorization of terminology, pathways, and classification systems while also requiring conceptual integration. This paper asks whether a structured spaced-repetition tool can improve both factual recall and conceptual understanding.

2. Methods

Participants were randomly assigned to one of two conditions. The experimental group used a flashcard application that scheduled review at expanding intervals of one, three, seven, and fourteen days. The control group was instructed to study the same material in a single weekly block. Both groups received identical lectures, textbooks, and total study time targets of roughly three hours per week.

Assessments combined multiple-choice recall items with short-answer questions requiring application of concepts to novel scenarios. Scores were analyzed using mixed-effects regression, controlling for prior GPA and baseline pretest performance.

3. Results

At the one-week assessment, the two groups performed similarly, with the control group scoring slightly higher on recall items (72% vs. 70%). By week four, the spaced-repetition group had pulled ahead (68% vs. 57%), and at week twelve the gap widened further (64% vs. 41%). The advantage was largest for terminology-heavy topics such as cellular respiration and taxonomy.

Improvements on conceptual short-answer questions were smaller but still significant ($p < 0.01$). Students in the experimental group also reported lower exam anxiety and greater confidence, though self-reported data should be interpreted cautiously.

4. Discussion and Limitations

The findings support integrating spaced-repetition tools into undergraduate science instruction, particularly in courses where cumulative knowledge is essential for later coursework. However, the study was limited to a single institution and a single subject area, and adherence to the review schedule was self-monitored. Future work should test whether benefits generalize to chemistry, physics, and non-STEM subjects, and whether instructor-led reminders improve adherence.

5. Conclusion

Spaced repetition produced substantially better long-term retention than massed practice at no additional time cost. Universities should consider embedding spacing-based study tools into learning management systems and training students in effective study habits during orientation.